`SpinOctopus`: Configuration Guide and Technical Documentation

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1 Introduction to the Framework

`SpinOctopus` is an integrated general-relativistic ray-tracing framework developed in Fortran and Python, specifically designed for axisymmetric spacetimes. By simply inputting the covariant metric of the target spacetime and the task configuration parameters into a dedicated script, users can efficiently execute simulations, including black hole image and light curve. For comprehensive technical details, please refer to the inline code documentation and the associated preprint available on arXiv: Shiyang Hu, Dan Li, Chen Deng, Lina Zhang, and Wenfu Cao, *SpinOctopus: A Versatile, User-Friendly, and Extensible Public Code for General-Relativistic Ray-Tracing in axisymmetric spacetime.*

2 Code Structure and Execution Workflow

`SpinOctopus` utilizes OpenMP multi-threading to accelerate the ray-tracing algorithm. The codebase comprises 8 Fortran source files, 10 Python scripts, and a `Makefile`. Notably, the Python script `spacetime_model.py` is utilized to define the target spacetime model and configure task parameters; it subsequently generates the `model.f90` module for Fortran compilation. All remaining Python scripts are dedicated to data visualization.

To execute `SpinOctopus`, all aforementioned files must be placed within a single root directory. Additionally, two subdirectories named `data` and `results` must be created within this root directory to store intermediate data and final outputs, respectively. Once the covariant metric components and task parameters are configured in `spacetime_model.py` and `model.f90` is generated, the Fortran source code can be compiled (using `gfortran` is highly recommended) to initiate the simulation. Result visualization is subsequently performed using the specific Python script corresponding to the executed task mode.

3 Code and Theoretical Formulation Correspondence

The SpinOctopus framework incorporates an extensive array of configuration parameters. To facilitate accurate setup and modifications, the direct correspondence between the underlying code variables, mathematical symbols, and the governing equations presented in the manuscript is provided in Tables 1—8.

4 Visualization

SpinOctopus utilizes dedicated Python scripts to visualize the output data generated across various task modes. A comprehensive summary detailing the specific task indices, generated data files, and their corresponding plotting scripts is provided in Table 9. In addition, during visualization, it is imperative to ensure that the plotting resolution strictly matches the computational resolution configured during the simulation execution.

Table 1: Correspondence between underlying code parameters, manuscript symbols, and equations.

Code Parameter	**Symbol**	**First Eq.**	**Description**
`t,r,theta,phi`	t, r, θ, φ	(2.1)	Generalized coordinates of the target metric.
`spin`	a	(2.21)	Dimensionless spin parameter.
`task_model`	N/A	N/A	Specifies the task execution mode.
`cpu`	N/A	N/A	Number of parallel threads for task execution.
`r_obs`	r_{obs}	(2.20)	Observation distance (typically fixed at 1000).
`theta_obs`	Θ	(2.20)	Observation inclination angle.
`phi_obs`	Φ	(2.18)	Observation azimuthal angle.
`resolution_x`	N/A	N/A	Horizontal resolution of the observation screen.
`resolution_y`	N/A	N/A	Vertical resolution of the observation screen.
`x_ini, x_end`	N/A	N/A	Horizontal dimensions of the field of view (FOV).
`y_ini, y_end`	N/A	N/A	Vertical dimensions of the field of view (FOV).
`ini_step`	$h_{\min}$	(2.17)	Minimum initial iteration step size for the RKF56 solver.
`step_ratio`	N	(2.17)	Step size amplification exponent.
`theta_correction`	ξ_θ	(2.17)	Step size control factor near the polar axis.
`j0`	j_0	(3.9)	Weight factor of the accretion environment emission.
`a0`	α_0	(3.14)	Weight factor of the accretion environment absorption.

Table 2: Correspondence between code parameters and manuscript symbols for the geometrically thick accretion environment.

Code Parameter	**Symbol**	**First Eq.**	**Description**
`r_inner`	$r_{\rm in}$	(3.10)	Inner boundary radius of the accretion environment.
`r_outer`	$r_{\rm out}$	N/A	Outer boundary radius of the accretion environment.
`disk_model`	N/A	N/A	Toggle flag determining the activation of `r_inner`.
`disk_intensity`	j_1	(3.9)	Weight factor for the geometrically thick disk emission.
`thick_emit_decay1`	p_1	(3.10)	Primary radial decay exponent of the geometrically thick disk.
`thick_emit_decay2`	p_2	(3.10)	Secondary radial decay exponent of the geometrically thick disk.
`thickness`	$\sigma_{d\theta}$	(3.10)	Initial thickness parameter of the geometrically thick disk.
`disk_extend`	δ_d	(3.10)	Thickness expansion factor of the geometrically thick disk.
`platform_intensity`	$j_{\rm p}$	(3.11)	Weight factor for the emission compensation.
`platform_location`	$r_{\rm p}$	(3.11)	Radial position of the emission compensation.
`platform_width`	$w_{\rm p}$	(3.11)	Width of the emission compensation.
`bump_intensity`	j_2	(3.9)	Weight factor for the Gaussian bump emission.
`bump_location`	$r_{\rm b}$	(3.12)	Radial position of the Gaussian bump.
`sigma_bump_r`	σ_{br}	(3.12)	Radial extent (standard deviation) of the Gaussian bump.
`sigma_bump_theta`	$\sigma_{b\theta}$	(3.12)	Polar extent (standard deviation) of the Gaussian bump.
`blob_intensity`	j_3	(3.9)	Weight factor for the local hotspot emission.
`blob_(r,theta,phi)`	$r_{\rm s}$, $\theta_{\rm s}$, $\varphi_{\rm s}$	(3.13)	Spatial coordinates of the local hotspot.
`sigma_blob_r`	σ_{sr}	(3.13)	Radial extent of the local hotspot.
`sigma_blob_theta`	$\sigma_{s\theta}$	(3.13)	Polar extent of the local hotspot.
`sigma_blob_phi`	$\sigma_{s\varphi}$	(3.13)	Azimuthal extent of the local hotspot.
`inspect_azimuth`	N/A	N/A	Azimuthal angle for inspecting the 1D emission profile of the accretion environment.

Table 3: Correspondence between code parameters and manuscript symbols for the accretion flow kinematics and geometrically thin disk models. Note that both geometrically thick and thin disks share the same kinematic framework, with the latter restricting particle motion strictly to the equatorial plane.

Code Parameter	**Symbol**	**First Eq.**	**Description**
`velocity_max_r`	$V^r_{\max}$	(3.15)	Maximum radial velocity of the accretion flow.
`velocity_max_theta`	$V^\theta_{\max}$	(3.16)	Maximum polar velocity of the accretion flow.
`velocity_max_phi`	$V^\varphi_{\max}$	(3.17)	Maximum azimuthal velocity of the accretion flow.
`velocity_decay_r`	p_3	(3.15)	Radial acceleration profile factor.
`velocity_decay_theta`	p_4	(3.16)	Polar acceleration profile factor.
`velocity_decay_phi`	p_5	(3.17)	Azimuthal acceleration profile factor.
`fudge_factor`	f_n	(3.33)	Fudge factor for fine-tuning rendered image details.
`thin_emit_decay1`	p_1	(3.30)	Primary radial decay exponent for the geometrically thin disk emission.
`thin_emit_decay2`	p_2	(3.30)	Secondary radial decay exponent for the geometrically thin disk emission.
`max_hit`	$\mathcal{N}$	(3.33)	Maximum number of times a ray is permitted to intersect the thin accretion disk.

Table 4: Correspondence between code parameters and manuscript symbols for the point source model.

Code Parameter	**Symbol**	**First Eq.**	**Description**
`point_source_(x,y,z)`	$x'_{\rm s}$, $y'_{\rm s}$, $z'_{\rm s}$	(3.45)	Spatial coordinates of the point source.
`point_source_radius`	$r_{\rm source}$	(3.44)	Radius of the point source.
`point_source_sigma`	$\sigma_{\rm point}$	(3.45)	Standard deviation of the point source emission profile.
`point_source_vt`	N/A	N/A	Instantaneous four-velocity components of the point source.
`point_source_vr`	N/A	N/A	Instantaneous four-velocity components of the point source.
`point_source_vtheta`	N/A	N/A	Instantaneous four-velocity components of the point source.
`point_source_vphi`	N/A	N/A	Instantaneous four-velocity components of the point source.

Table 5: Correspondence between code parameters and manuscript symbols for the jet outflow model.

Code Parameter	Symbol	First Eq.	Description
`jet_intensity`	j_0	(3.38)	Intensity weighting factor for the jet emission.
`jet_base`	$r_{\rm base}$	(3.39)	Radial coordinate of the jet base.
`jet_range`	N/A	N/A	Artificially imposed truncation radius of the jet.
`jet_decay`	p_7	(3.39)	Radial decay exponent for the jet emission.
`jet_angle_ini`	$\vartheta_{\rm base}$	(3.41)	Initial opening angle of the jet.
`jet_angle_inf`	ϑ_∞	(3.41)	Asymptotic (saturation) opening angle of the jet.
`jet_angle_expansion`	δ_j	(3.41)	Characteristic scale for the expansion of the jet.
`jet_velocity_ini`	$V_{\rm base}$	(3.42)	Initial radial velocity of the jet.
`jet_velocity_inf`	$V_{\rm max}^{\rm jet}$	(3.42)	Asymptotic (saturation) radial velocity of the jet.
`jet_omega`	$\omega_{\rm jet}$	(3.43)	Maximum azimuthal velocity of the particles within the jet.
`jet_omega_decay`	p_8	(3.43)	Decay exponent for the azimuthal velocity of the jet particles.
`gyration_radius`	$r_{\rm gyra}$	(3.43)	Minimum gyration radius of the particles within the jet.

Table 6: Correspondence between code parameters and manuscript symbols for the spherically symmetric accretion shell model.

Code Parameter	Symbol	First Eq.	Description
`spherically_flow_intensity`	j_0	(3.36)	Intensity weighting factor for the spherical shell emission.
`spherically_flow_outer`	$r_{\rm out}$	N/A	Outer boundary radius of the spherical shell.
`spherically_flow_inner`	$r_{\rm in}$	N/A	Inner boundary radius of the spherical shell.
`spherically_flow_peak`	$r_{\rm shell}$	(3.35)	Radial position of the peak emission within the shell.
`infalling_flow_max`	$V_{\rm max}^{\rm shell}$	(3.37)	Maximum radial velocity of the infalling accretion flow.
`infalling_accelerating`	p_6	(3.37)	Radial acceleration profile factor for the infalling flow.
`emission_brake`	$\delta_{\rm brake}$	(3.37)	Braking coefficient for the infalling matter crossing the shell.
`spherically_flow_sigma`	$\sigma_{\rm shell}$	(3.35)	Radial extent (standard deviation) of the shell emission.

Table 7: Correspondence between code parameters and manuscript symbols for the timelike particle trajectory simulation.

Code Parameter	**Symbol**	**First Eq.**	**Description**
`(t,r,theta,phi)_mp`	t, r, θ, φ	(2.1)	Generalized coordinates of the particle.
`(pt,pr,ptheta,pphi)_mp`	p_t, p_r, p_θ, p_φ	(2.4)—(2.7)	Conjugate momenta of the particle.
`step_mp`	N/A	N/A	Integration step-size for the timelike trajectory.
`motion_time`	N/A	N/A	Total proper time for the particle's motion.
`max_point`	N/A	N/A	Maximum number of data points recorded for the trajectory.
`sample`	N/A	N/A	Sampling frequency for trajectory data export.
`particle_radius`	r_{source}	(3.46)	Radius of the discretized spheres used to render the particle trajectory.
`condition_type`	N/A	N/A	Method parameter used to compute the initial conditions of the particle.

Table 8: Correspondence between code parameters and manuscript symbols for the remaining configuration settings not covered in previous tables.

Code Parameter	**Symbol**	**First Eq.**	**Description**
`disk_inclination`	χ_{t}	(3.34)	Tilt angle of the geometrically thin tilted accretion disk.
`(x,y)_start`	x, y	(2.18), (2.20)	Initial pixel coordinates on the observation screen for single ray-tracing.
`show_line`	N/A	N/A	Toggle flag for rendering the celestial sphere grid lines (longitude and latitude).
`delta_latitude`	N/A	N/A	Latitudinal interval spacing for the celestial background source grid.
`delta_longitude`	N/A	N/A	Longitudinal interval spacing for the celestial background source grid.

Table 9: Correspondence between task modes, generated output files, visualization plot types, and plotting scripts.

Task Index	**Task Description**	**Generated Data**	**Plot Type**	**Plotting Script**
1	Critical curve (Shadow)	`shadow.dat`	Scatter	`shadow_profile_plot.py`
2	Thick disk emission	`j-profile-r(roz,xoy).dat`	Contour, Line	`disk_emission_plot.py`
3	Thick disk image	`image.dat`	Contour	`image_plot.py`
4	Thin disk image	`image.dat`, `redshift-1(2,3,4).dat`, `thin-disk-hit.dat`	Contour	`image_plot.py`, `redshift_plot.py`, `hit_plot.py`
5	Four-color image	`four-color-image.dat`	Contour	`four_color_plot.py`
6	Jet emission	`j-profile-roz.dat`	Contour	`jet_emission_plot.py`
7	Jet image	`image.dat`	Contour	`image_plot.py`
8	Ray trajectory	`particle-path.dat`, `energy-error.dat`	Line	`path_plot.py`
9	Timelike trajectory	`particle-path.dat`, `energy-error.dat`	Line	`path_plot.py`
10	Point source image	`image.dat`	Contour	`image_plot.py`
11	Light curve	`time-series.dat`	Line	`light_curve_plot.py`
12	Spherical shell image	`image.dat`	Contour	`image_plot.py`
13	Celestial grid image	`grid.dat`	Scatter	N/A
14	Tilted thin disk image	`four-color-image.dat`	Contour	`four_color_plot.py`